Keyvan M. Sadeghi

Senior AI Solutions Architect

linkedin.com/in/keyvanmsadeghi github.com/keyvan-m-sadeghi keyvan.m.sadeghi@gmail.com +1 416-859-8984
Toronto, Ontario, Canada Open to Relocate/Remote

SUMMARY

Results-oriented AI/ML leader & architect with 10+ years of experience in designing, building, and managing the end-to-end delivery of complex AI solutions (LLMs, RAG, Agents) and distributed systems. Proven expertise in project planning, execution, risk mitigation, and process improvement within agile frameworks, supporting engineering and product teams to deliver high-performance computing and AI libraries. Adept at translating complex requirements into technical strategy and actionable plans, fostering collaboration across geographically distributed teams, and leveraging data analysis to drive operational efficiencies. Expertise in AI/ML system architecture, MLOps, cloud infrastructure, and AI platform optimization.

EDUCATION

University of Southampton, UK 2010 – 2012
MSc, Artificial Intelligence – Grade: Distinction

EXPERIENCE

Co-founder, CEO / CSO *Jan 2021 – Present*
Functionland *Toronto, ON*

- Architected decentralized AI solutions; deployed fine-tuned LLMs (DeepSeek R1) on edge devices (RK3588/TPU) across 1k+ deployed nodes (1PB+ storage).
- Developed RAG systems & agentic frameworks, managing stakeholder relationships with partners (Fetch.ai) & enterprise clients; co-designed distinctive hardware (FxBlox) with Yves Béhar enabling versatile edge computing use cases.
- Led technical strategy, managed budgets, grew and managed core/contractor team (to 70+ at peak), scaled production infra; secured $500k+ crowdfunding & $1.1M seed round; deployed 3 consumer apps (Google Play/Apple App Store).
- Established CI/CD (GitHub Actions), MLOps processes, monitoring, and system maintenance for AI workloads.

Principal Software Engineer *Dec 2018 – Aug 2021*
Assister Technology Ltd *Toronto, ON*

- Revamped client applications with React/Native front-ends, Node.js/Go back-ends, and NoSQL/SQL databases.
- Led design/development of Python-based conversational AI platform (Transformer models, PyTorch) for enterprise clients, incorporating formal project planning/tracking & managing stakeholder relations.
- Deployed scalable cloud-native apps (AWS) using CI/CD (GH Actions), driving operational process improvements.
- Provided architectural guidance for AI solutions; chaired W3C FKG group, drafting specs for web-Agent communication.

Lead AI Architect *Oct 2014 – Nov 2018*
Mahan Air, MAPS Ventures *Tehran, Iran*

- Scaled and managed AI/Analytics team (from 2 to over 20 engineers), implementing Agile methodologies (Scrum/Kanban), creating project status dashboards, and developing process documentation; established technical direction.
- Spearheaded project leveraging ML to automate fleet assignment, showcasing AI's potential in aviation operations.
- Applied Agile principles to shape architectures (incl. Spark, Neo4j) & tech stacks (Python, Node.js).

AGI Researcher *Feb 2012 – Sep 2014*
Hong Kong Polytechnic University *Hong Kong*

- Programmed AI Agents embodied in a Unity 4.0 game simulation, defining objective and memory for NPCs.
- Published four papers on cognitive architectures, advanced AI reasoning (spatio-temporal), and genetic algorithms.

SKILLS

Program & Project Management	Agile (Scrum, Kanban), Project Planning & Execution, End-to-End Delivery, Risk Management, Stakeholder Management, Process Improvement, Resource Planning, Operational Efficiency, Data Analysis & Presentation, Reporting & Forecasting
PM Tools	Jira, Confluence, Microsoft Excel (Expert), Microsoft Project (familiarity), Data Visualization (e.g., Power BI concepts), Git, GitHub
AI / ML	LLMs (Deployment, RAG, Agents), NLP, Transformers, PyTorch, MLOps, Model Evaluation, High-Performance Computing (HPC) concepts, GPU Software Development concepts, AI/Math/Computer Vision Library concepts
Cloud & DevOps	AWS, GCP, Azure (familiarity), Kubernetes, Docker, CI/CD (GitHub Actions, Azure DevOps), Distributed Systems, Performance Tuning, Monitoring, IaC concepts
Software Dev & Architecture	Python (Expert), Node.js, JavaScript/TypeScript, System Design, Microservices
Leadership & Comms	Team Leadership, Cross-Functional Collaboration, Mentorship, Excellent Written/Verbal Communication, Conflict Resolution, Training Material Development

PUBLICATIONS / REFERENCES

Papers accessible from this link. Reference letters available for download at this link.